Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 1 (Generated practice paper - Set 33)

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
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9	
10	
11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 . 1 Which of the following best describes the purpose of Critical Group Monitoring? Tick one box.

[1 mark]

A To detect health damage in the population most affected by a pollutant ☐

B To predict the route a pollutant will take through the environment ☐

C To assess exposure of the people most at risk so that everyone else can be assumed to be sa ☐

D To measure the concentration of a pollutant at the point of release ☐

0 1 . 2 Explain why CPA can reduce the cost of an environmental monitoring programme.

[2 marks]

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0 1 . 3 A new chemical has properties similar to PCBs (persistent, liposoluble chlorinated organic compounds). Explain how knowledge of PCBs helps decisions about the new chemical.

[2 marks]

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0 2 . 1 Describe how ozone is formed and destroyed naturally in the stratosphere.

[3 marks]

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0 2 . 2 Explain the properties of CFCs that made them useful, and the properties that cause ozone depletion (the Rowland-Molina hypothesis).

[6 marks]

Useful properties

Properties causing ozone depletion

0 2 . 3 Explain why increased UVB reaching the Earth's surface is harmful to living organisms. Give **three** effects.

[3 marks]

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0 2 . 4 State what is meant by an **ozone hole**.

[1 mark]

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0 2 . 5 Give **one** reason why concentration in ppm is not used to measure ozone depletion.

[1 mark]

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0 2 . 6 Give **one** reason why the first evidence for ozone depletion came from ground-based measurements.

[1 mark]

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0 3 . 1 Explain how each of the following features of vehicle design reduces energy use.

[6 marks]

Aerodynamics .....

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Reduced mass .....

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Regenerative braking (KERS) .....

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0 3 . 2 Explain why solid steel wheels on rails lose less energy than pneumatic tyres on roads.

[2 marks]

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0 3 . 3 Describe **two** ways combustion efficiency in an internal combustion engine can be improved.

[2 marks]

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0 4 . 1 Describe the two processes that drive the North Atlantic Conveyor.

[2 marks]

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0 4 . 2 Explain how global climate change could weaken the North Atlantic Conveyor and describe the possible consequence for the climate of the UK.

[4 marks]

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0 4 . 3 Explain why the effects of climate change on UK temperatures are difficult to predict.

[3 marks]

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0 4 . 4 Describe how deep ocean currents are monitored.

[1 mark]

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0 5 Figure 1 shows the daily deaths in London and the mean concentration of smoke in the air during the smog of December 1952.

Figure 1

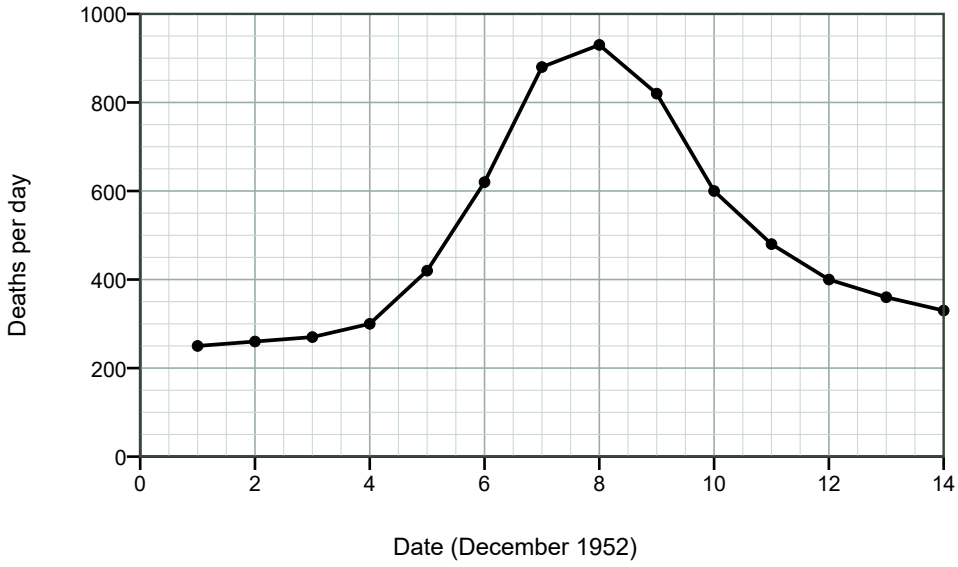


Table 1

Date	3 Dec	5 Dec	7 Dec	8 Dec	10 Dec	12 Dec
Smoke / mg m^{-3}	0.4	1.5	4.4	4.5	1.2	0.5

0 5 . 1 Describe the relationship between smoke concentration and daily deaths shown in Figure 1 and Table 1.

[2 marks]

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0 5 . 2 Calculate the percentage increase in daily deaths between 1 December and 8 December.

[2 marks]

Answer ..... %

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0 5 . 3 Explain how the weather conditions in December 1952 caused the smog to form.

[3 marks]

0 5 . 4 Explain why most of the deaths were among the elderly, the very young and people with existing respiratory disease.

[2 marks]

0 5 . 5 State the legislation introduced in the UK in response to this smog.

[1 mark]

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0 6 . 1 Describe how DNA databases are used to control the illegal trade in tropical timber and ivory.

[3 marks]

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0 6 . 2 Explain the difference between radio / GPS tracking and acoustic tracking and state when each is used.

[2 marks]

0 6 . 3 Give **three** types of carrier system used to move sensors to study areas and one advantage of drones for ecological monitoring.

[3 marks]

0 6 . 4 Explain how motion-sensitive cameras and CCTV reduce disturbance to wildlife compared with direct observation.

[2 marks]

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0 7 . 1 Which of the following is a **secondary** pollutant? Tick one box.

[1 mark]

A Sulfur dioxide from a coal-fired power station ☐

B Tropospheric ozone ☐

C Carbon monoxide from a car exhaust ☐

D Nitrogen monoxide from a diesel engine ☐

0 7 . 2 Explain the difference between a primary pollutant and a secondary pollutant, using acid rain as an example.

[4 marks]

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0 8 An aquifer beneath an agricultural plain holds 3.6×10^{11} m^3 of water. It is recharged at 1.2×10^9 m^3 per year, and 4.8×10^9 m^3 per year is abstracted from it.

0 8 . 1 Residence time = volume of the reservoir / rate of flow through it. Calculate the residence time of the water in this aquifer, using the recharge rate.

[2 marks]

Answer ☐ years

0 8 . 2 Explain the properties a rock must have to form a useful aquifer.

[3 marks]

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0 8 . 3 Explain why the water table beneath the plain is falling.

[2 marks]

0 8 . 4 Describe **four** consequences of continuing to over-abstract this aquifer.

[4 marks]

0 8 . 5 Explain why an aquifer holding 'fossil water' cannot be exploited sustainably, and describe **two** measures that would reduce abstraction.

[4 marks]

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0 9 A student measured the mass of carbon stored per hectare in three land uses in the same region.

Table 1

Land use	Carbon in vegetation / t ha^{-1}	Carbon in soil dead organic matter / t ha^{-1}
Mature woodland	120	95
Permanent pasture	3	80
Arable (ploughed annually)	2	45

0 9 . 1 Calculate the total carbon lost per hectare when mature woodland is converted to arable farmland.

[1 mark]

Answer ☐ t/ha

0 9 . 2 Explain why the soil carbon in arable land is so much lower than in pasture, even though the vegetation carbon is similar.

[2 marks]

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0 9 . 3 Suggest **two** ways the arable farmer could increase soil carbon storage.

[2 marks]

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1 0 . 1 Evaluate the risks and benefits of the uses of ionising radiation, and explain the principles used to control exposure of workers and the public.

[9 marks]

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Extra space

1 0 . 2 Give **one** reason why the risks and benefits of nuclear electricity may not fall on the same people.

[1 mark]

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Write an essay on one of the following topics.

1 1 . 1 Discuss the extent to which the study of ozone depletion illustrates how scientific investigation can lead to the successful solution of an environmental problem.

[25 marks]

OR

1 1 . 2 Discuss the extent to which energy conservation technologies could reduce the need for new energy supplies in a developed country.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS